

Sur Protocol Whitepaper

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Sur Protocol

Abstract

Introduction

With the development of AI and as we are moving towards AGI, it is becoming increasingly important to know with whom we are interacting on the internet. In the upcoming era of AI, more and more AI agents will be interacting with each other and with humans. It is crucial to ensure that these interactions are secure, trustworthy, and verifiable. While protocols like DKG¹ provide a foundation for decentralized data origin and provenance technologies, there is a need to build a protocol adapted to latest LLM technologies and that ensures a better experience for users and AI agents.

The idea of the Sur Protocol comes from the question: “Can we ensure that produced data can be authenticated distinctively and transparently by interlocutors?”

We believe that in the coming days, it will be very difficult to differentiate impersonation data on the Internet.

¹DKG: Decentralized Knowledge Graph <https://dkg.io>

Despite having a writing style, a voice intonation, and a face, it will be easy for machines to be able to usurp anyone's identity: from a simple village inhabitant requesting a fraudulent divorce to the President of the Republic triggering a war.

The Sur Protocol aims to be a tool for humanity that connects the physical creation of data to highly decentralized verification systems to avoid any conflicts of interest.

The data generated each day is massive, the protocol must allow for ensuring the traceability of data while ensuring their confidentiality.

The Sur Protocol is based on the following technologies:

- Blockchain: for decentralization
- ZK: for generating proofs of data creation and non-disclosure of sensitive data
- Data provenance: for tracking data modifications that may occur

Use Cases

This protocol aims to be a tool for the following use cases:

- Authentication of the provenance of messages sent on social networks.
- Authentication of the provenance of official documents (birth certificates, diplomas, contracts, etc.).
- Authentication of video and audio calls (deepfake detection).
- Authentication of the provenance of images and videos (deepfake detection).
- Authentication of the provenance of data generated by AIs (text, image, video, audio, etc.).

Architecture

Tokenomics

Roadmap

Conclusion

References